

**LAPORAN PTAKTIKUM BASIS DATA
MEMBUAT DATABASE LSP MENGGUNAKAN MARIA DB**



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TAHUN AKADEMIK 2025/2026**

HASIL Pengerjaan

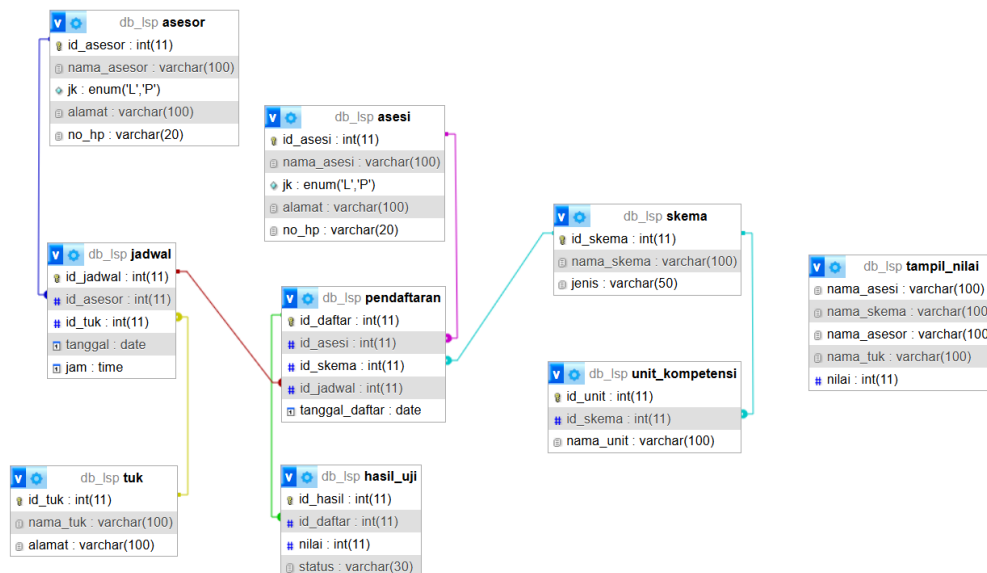
1. Database yang dibuat memiliki nama db_lsp dan terdiri dari delapan tabel, yaitu:

- asesor
- asesi
- skema
- unit_kompetensi
- tuk
- jadwal
- pendaftaran
- hasil_uji

Relasi Antar Tabel

- Asesor memiliki banyak Jadwal (1:N)
- TUK memiliki banyak Jadwal (1:N)
- Jadwal memiliki banyak Pendaftaran (1:N)
- Asesi memiliki banyak Pendaftaran (1:N)
- Skema memiliki banyak Unit Kompetensi (1:N)
- Skema memiliki banyak Pendaftaran (1:N)
- Pendaftaran memiliki satu Hasil Uji (1:1)

HASIL:



2. Implementasi Database

Database diimplementasikan menggunakan MariaDB dengan nama **db_lsp**.

```
Microsoft Windows [Version 10.0.26200.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\devia>cd C:\xampp\mysql\bin

C:\xampp\mysql\bin>mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 9
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE db_lsp;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> USE db_lsp;
Database changed
```

3. Pembuatan Tabel Menggunakan Command Line Interface (CLI)

Seluruh tabel dibuat menggunakan MariaDB Command Line Interface (CLI).

```
MariaDB [db_lsp]> SHOW DATABASES;
+-----+
| Database |
+-----+
| db_lsp   |
| db_perpus_digital |
| db_sidang_ta |
| hotel_db |
| hotel_devi |
| information_schema |
| mysql    |
| performance_schema |
| perpustakaan_digital |
| phpmyadmin |
| sekolah_sd |
| toko_baju |
| toko_baju_lengkap |
| tugas11_agregate_mumtaz |
+-----+
14 rows in set (0.083 sec)
```

4. Memasukkan Data (INSERT INTO)

Data berhasil dimasukkan ke dalam seluruh tabel sesuai ketentuan soal.

Proses pengisian data dilakukan menggunakan perintah **INSERT INTO** pada masing-masing tabel.

```

MariaDB [db_lsp]> INSERT INTO asesor(nama_asesor,jk,alamat,no_hp)
-> VALUES
-> ('Andi Saputra','L','Tegal','08111111111'),
-> ('Budi Santoso','L','Slawi','08111111112'),
-> ('Citra Lestari','P','Brebes','08111111113'),
-> ('Dewi Anggraini','P','Pemalang','08111111114'),
-> ('Eko Prasetyo','L','Purwokerto','08111111115');
Query OK, 5 rows affected (0.041 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO asesi(nama_asesi,jk,alamat,no_hp)
-> VALUES
-> ('Dewi Aliya','P','Slawi','081111111201'),
-> ('Mega','P','Balapulang','081111111202'),
-> ('Sekar','P','Kramat','081111111203'),
-> ('Mufida','P','Balapulang','081111111204'),
-> ('Ismi','P','Balapulang','081111111205'),
-> ('Lia','P','Balapulang','081111111206'),
-> ('Hanin','P','Tegal','081111111207'),
-> ('Amel','P','Tegal','081111111208'),
-> ('Kirana','P','Slawi','081111111209'),
-> ('Sabil','L','Tegal','081111111210'),
-> ('Fatkhil','L','Adiwerna','081111111211'),
-> ('Azam','L','Balamoa','081111111212'),
-> ('Firza','L','Slawi','081111111213'),
-> ('Izan','L','Bojong','081111111214'),
-> ('Asrop','L','Bojong','081111111215'),
-> ('Khirzi','L','Brebes','081111111216'),
-> ('Baim','L','Adiwerna','081111111217'),
-> ('Hanip','L','Kalisapu','081111111218'),
-> ('Dika','L','Brebes','081111111219'),
-> ('Nadia','P','Batu Agung','081111111220');
Query OK, 20 rows affected (0.008 sec)
Records: 20 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO skema(nama_skema,jenis)
-> VALUES
-> ('Junior Web Developer','IT'),
-> ('Junior Programmer','IT'),
-> ('Database Administrator','IT'),
-> ('Network Administrator','IT'),
-> ('Graphic Designer','Multimedia');
Query OK, 5 rows affected (0.007 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO unit_kompetensi(id_skema,nama_unit)
-> VALUES
-> (1,'HTML'),
-> (1,'CSS'),
-> (1,'JavaScript'),
-> (1,'PHP'),
-> (2,'Algoritma'),
-> (2,'Flowchart'),
-> (2,'OOP'),
-> (2,'Pemrograman'),
-> (3,'DDL'),
-> (3,'DML'),
-> (3,'View'),
-> (3,'Trigger'),
-> (4,'Routing'),
-> (4,'Switching'),
-> (4,'Firewall'),
-> (4,'Server'),
-> (5,'Photoshop'),
-> (5,'Illustrator'),
-> (5,'CorelDraw'),
-> (5,'Figma');
Query OK, 20 rows affected (0.013 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO tuk(nama_tuk,alamat)
-> VALUES
-> ('TUK A','Tegal'),
-> ('TUK B','Slawi'),
-> ('TUK C','Brebes'),
-> ('TUK D','Pemalang'),
-> ('TUK E','Purwokerto');
Query OK, 5 rows affected (0.007 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO jadwal(id_asesor,id_tuk,tanggal,jam)
-> VALUES
-> (1,1,'2026-07-28','08:00:00'),
-> (2,2,'2026-07-28','09:00:00'),
-> (3,3,'2026-07-29','08:00:00'),
-> (4,4,'2026-07-29','09:00:00'),
-> (5,5,'2026-07-30','08:00:00'),
-> (1,2,'2026-07-30','10:00:00'),
-> (2,3,'2026-07-31','08:00:00'),
-> (3,4,'2026-07-31','09:00:00'),
-> (4,5,'2026-08-01','08:00:00'),
-> (5,1,'2026-08-01','10:00:00');
Query OK, 10 rows affected (0.021 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO pendaftaran(id_asesi,id_skema,id_jadwal,tanggal_daftar)
-> VALUES
-> (1,1,1,'2026-07-20'),
-> (2,2,2,'2026-07-20'),
-> (3,3,3,'2026-07-20'),
-> (4,4,4,'2026-07-20'),
-> (5,5,5,'2026-07-20'),
-> (6,1,6,'2026-07-20'),
-> (7,2,7,'2026-07-20'),
-> (8,3,8,'2026-07-20'),
-> (9,4,9,'2026-07-20'),
-> (10,5,10,'2026-07-20'),
-> (11,1,1,'2026-07-21'),
-> (12,2,2,'2026-07-21'),
-> (13,3,3,'2026-07-21'),
-> (14,4,4,'2026-07-21'),
-> (15,5,5,'2026-07-21'),
-> (16,1,6,'2026-07-21'),
-> (17,2,7,'2026-07-21'),
-> (18,3,8,'2026-07-21'),
-> (19,4,9,'2026-07-21'),
-> (20,5,10,'2026-07-21');
Query OK, 20 rows affected (0.072 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO hasil_uji(id_daftar,nilai,status)
-> VALUES
-> (1,90,'Kompeten'),
-> (2,85,'Kompeten'),
-> (3,75,'Kompeten'),
-> (4,65,'Belum Kompeten'),
-> (5,80,'Kompeten'),
-> (6,92,'Kompeten'),
-> (7,88,'Kompeten'),
-> (8,70,'Kompeten'),
-> (9,60,'Belum Kompeten'),
-> (10,95,'Kompeten'),
-> (11,89,'Kompeten'),
-> (12,78,'Kompeten'),
-> (13,83,'Kompeten'),
-> (14,68,'Belum Kompeten'),
-> (15,91,'Kompeten'),
-> (16,77,'Kompeten'),
-> (17,84,'Kompeten'),
-> (18,86,'Kompeten'),
-> (19,72,'Kompeten'),
-> (20,93,'Kompeten');
Query OK, 20 rows affected (0.023 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> CREATE VIEW tampil_nilai AS
-> SELECT
-> a.nama_asesi,
-> s.nama_skema,
-> asr.nama_asesor,
-> t.nama_tuk,
-> h.nilai
-> FROM hasil_uji h
-> JOIN pendaftaran p ON h.id_daftar=p.id_daftar
-> JOIN asesinya a ON p.id_asesi=a.id_asesi
-> JOIN skema s ON p.id_skema=s.id_skema
-> JOIN jadwal j ON p.id_jadwal=j.id_jadwal
-> JOIN asesor asr ON j.id_asesor=asr.id_asesor
-> JOIN tuk t ON j.id_tuk=t.id_tuk;
Query OK, 0 rows affected (0.022 sec)

```

5. Pembuatan VIEW

VIEW dibuat dengan nama **tampil_nilai**.

VIEW digunakan untuk menampilkan informasi dari beberapa tabel yang telah di-*join*.

Data yang ditampilkan meliputi:

- Nama Asesi
- Nama Skema
- Nama Asesor
- Nama TUK
- Nilai

```
MariaDB [db_lsp]> CREATE VIEW tampil_nilai AS
-> SELECT
-> a.nama_asesi,
-> s.nama_skema,
-> asr.nama_asesor,
-> t.nama_tuk,
-> h.nilai
-> FROM hasil_uji h
-> JOIN pendaftaran p ON h.id_daftar=p.id_daftar
-> JOIN asesi a ON p.id_asesi=a.id_asesi
-> JOIN skema s ON p.id_skema=s.id_skema
-> JOIN jadwal j ON p.id_jadwal=j.id_jadwal
-> JOIN asesor asr ON j.id_asesor=asr.id_asesor
-> JOIN tuk t ON j.id_tuk=t.id_tuk;
Query OK, 0 rows affected (0.022 sec)

MariaDB [db_lsp]> SELECT * FROM tampil_nilai;
```

nama_asesi	nama_skema	nama_asesor	nama_tuk	nilai
Devi Aliya	Junior Web Developer	Andi Saputra	TUK A	90
Lia	Junior Web Developer	Andi Saputra	TUK B	92
Fatkhi	Junior Web Developer	Andi Saputra	TUK A	89
Khirzi	Junior Web Developer	Andi Saputra	TUK B	77
Mega	Junior Programmer	Budi Santoso	TUK B	85
Hanin	Junior Programmer	Budi Santoso	TUK C	88
Azam	Junior Programmer	Budi Santoso	TUK B	78
Baim	Junior Programmer	Budi Santoso	TUK C	84
Sekar	Database Administrator	Citra Lestari	TUK C	75
Amel	Database Administrator	Citra Lestari	TUK D	70
Firza	Database Administrator	Citra Lestari	TUK C	83
Hanip	Database Administrator	Citra Lestari	TUK D	86
Mufida	Network Administrator	Dewi Anggraini	TUK D	65
Kirana	Network Administrator	Dewi Anggraini	TUK E	60
Izan	Network Administrator	Dewi Anggraini	TUK D	68
Dika	Network Administrator	Dewi Anggraini	TUK E	72
Ismi	Graphic Designer	Eko Prasetyo	TUK E	80
Sabil	Graphic Designer	Eko Prasetyo	TUK A	95
Asrop	Graphic Designer	Eko Prasetyo	TUK E	91
Nadia	Graphic Designer	Eko Prasetyo	TUK A	93

```
20 rows in set (0.010 sec)
```

6. Pembuatan Subquery

a. Menampilkan Peserta dengan Nilai di Atas Rata-rata

Subquery digunakan untuk menampilkan peserta yang memperoleh nilai lebih tinggi dari rata-rata seluruh peserta.

Hasil:

```
MariaDB [db_lsp]> SELECT nama_asesi,nilai
-> FROM asesi
-> JOIN pendaftaran USING(id_asesi)
-> JOIN hasil_uji USING(id_daftar)
-> WHERE nilai >
-> (
-> SELECT AVG(nilai)
-> FROM hasil_uji
-> );
```

nama_asesi	nilai
Devi Aliya	90
Mega	85
Lia	92
Hanin	88
Sabil	95
Fatkhi	89
Firza	83
Asrop	91
Baim	84
Hanip	86
Nadia	93

```
11 rows in set (0.027 sec)
```

b. Menampilkan Asesor yang Belum Memiliki Jadwal

Subquery digunakan untuk mengetahui asesor yang belum mendapatkan jadwal sertifikasi.

Hasil:

```
MariaDB [db_lsp]> SELECT *
-> FROM asesor
-> WHERE id_asesor NOT IN
-> (
-> SELECT id_asesor
-> FROM jadwal
-> );
Empty set (0.016 sec)
```

c. Menampilkan Skema dengan Peserta Terbanyak

Subquery digunakan untuk mengetahui skema sertifikasi yang memiliki jumlah peserta paling banyak.

Hasil:

```
MariaDB [db_lsp]> SELECT nama_skema
-> FROM skema
-> WHERE id_skema =
-> (
-> SELECT id_skema
-> FROM pendaftaran
-> GROUP BY id_skema
-> ORDER BY COUNT(*) DESC
-> LIMIT 1
-> );
+-----+
| nama_skema |
+-----+
| Junior Web Developer |
+-----+
1 row in set (0.007 sec)
```